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ORIGINAL RESEARCH

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Disabled throwing shoulder/elbow players have high rates of impaired foot function

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ABSTRACT

The prevalence of impaired foot function and floating toes of 91 baseball players with disabled throwing shoulder or elbow were retrospectively investigated. Foot function was evaluated by foot "rock paper scissors" and floating toes were confirmed if none of the toes made contact with the mat in standing posture. The prevalence of impaired foot function and floating toes and the relationship between between them were determined. Abnormal foot function was observed in 83 players (91%); of those, 73 players (88%) also had floating toes, and the prevalence was statistically significant compared to those without it (12%) ($P < 0.001$). Floating toes were observed in 74 players (81%); of those, 73 players (99%) had impaired foot function, and the prevalence was also statistically significant compared to those without them (59%) ($P < 0.001$). Baseball players with disabled throwing shoulder or elbow have high rates of impaired foot function and floating toes.

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KEYWORDS
Disabled throwing shoulder; disabled throwing elbow; foot function; floating toe; kinetic chain

Introduction

Pitching motion is an integrated motion of the entire body, from the lower extremity to the upper extremity, needed to maximize rapid ball release (Seroyer et al., 2010). To understand pitching motion, recognizing the kinetic chain is undoubtedly important, which is a mechanical linkage of the body segments from the lower extremity to the upper extremity (Kibler et al., 2013). Ideal anatomy, physiology, and mechanics are required throughout all the body segments involved (Kibler et al., 2013). However, deterioration of these mechanics and functions results in breakdown of the kinetic chain, as it reduces its efficiency of energy transfer, consequently increasing mechanical stress to the shoulder and elbow, which induces disabled throwing injuries (Seroyer et al., 2010). Many factors regarding breakdown of the kinetic chain have been reported previously: scapular dyskinesis or dysfunction (Burkhart, Morgan, Kibler et al., 2003b), rotator cuff weakness (Burkhart, Morgan, Kibler et al., 2003b), loss of glenohumeral internal rotation (Wilk et al., 2011), limitation of hip internal rotation (Robb et al., 2010), and decreased balance (Garrison et al., 2013). To perform pitching motion

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through proper kinetic chain, the motion should start from the proximal segment and progress to the distal segments (Burkhart, Morgan, Kibler et al., 2003a, 2003b). The lower extremity, representing the proximal segment during pitching motion, serves as the main force generator of the kinetic chain (Burkhart, Morgan, Kibler et al., 2003a), and, it is recognized as an integral part of the pitching motion through its importance as a stable base (Burkhart, Morgan, Kibler et al., 2003a; Seroyer et al., 2010). Alterations in lower extremity control during the throwing motion may alter the location of the shoulder and elbow, potentially resulting in increased stress across both joints (Hannon et al., 2014). Although the foot is the most distal segment in the lower extremity and a relatively small base of support on which the body maintains balance, minor biomechanical alterations of its functions may influence the postural-control strategies (Cole et al., 2005). However, despite the importance of the foot and its function, there are no reports on foot function or floating toes among the players with disabled throwing shoulders and elbows. Therefore, the purpose of this study was to examine the relationship between foot function and floating toes among players with disabled throwing shoulders and elbows. We hypothesized that the players with a disabled throwing shoulder and elbow demonstrate a high rate of impairment in foot function and floating toes.

Methods

We designed this study as a retrospective observational study which was approved by the institutional review board of our institution (IRB number 2–2). All procedures for this study were conducted in accordance with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. We included a total of 167 baseball players who visited our outpatient clinic between April 2015 and March 2019 and were diagnosed with a disabled throwing shoulder or elbow. The diagnosis of the disabled throwing shoulder or elbow was made by the following criterion; players complained of shoulder or elbow pain during throwing, or players who were able to induce shoulder or elbow pain during physical examination, such as positive subacromial impingement sign or moving valgus stress test. We excluded participants if we could not evaluate foot function or if they had a history of foot injury (i.e., ankle sprain or fracture). All participants underwent physical examination by the senior author, followed by ultrasonography, plain radiography, computed tomography, or magnetic resonance imaging of the shoulder or elbow, if needed. We evaluated foot function by foot "rock paper scissors" movements according to previous report (Hasegawa et al., 2006). Players lay on a table relaxed. They were instructed to flex all the toes to create "rock" with ankle in plantar flexion position followed by extending the hallux while keeping other toes flexed to create "scissors". The evaluation was completed with the instruction to extend and spread all the toes to create "papers" (Figure 1). If the player was unable to perform one of the three movements, foot function was classified as impaired (Hasegawa et al., 2006). In addition to foot "rock paper scissors," we also evaluated floating toes (Figure 2; Fukuyama & Maruyama, 2011; Fukuyama et al., 2009; Matsuda et al., 2013; Saito et al., 2019). The players stood on a solid mat on both feet, postured in static upright position. They kept their feet at shoulder width while gazing at a marker 2 m ahead at eye level. We defined floating toes, if none of the toes made contact with the mat in the standing posture (Fukuyama & Maruyama, 2011). To

Figure 1. Definition of foot "rock paper scissors" movements. (a) "rock," (b) "scissors," and (c) "papers.".

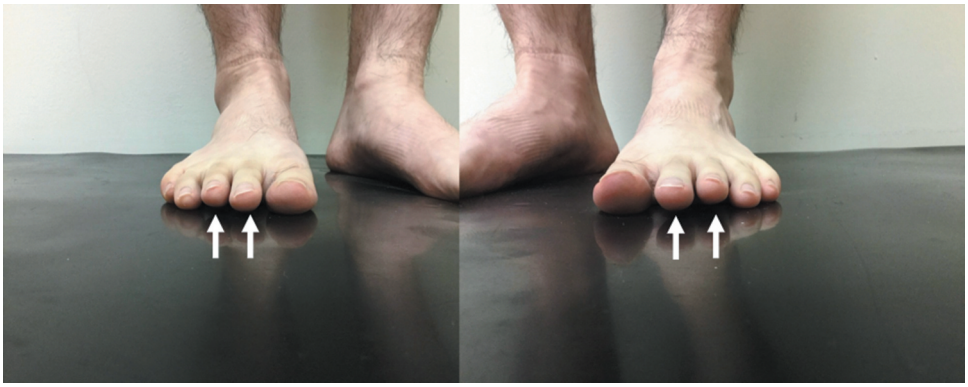

Figure 2. Characteristic image of floating toes. Arrows indicate toes that are floated.

eliminate inter-observer error, the evaluation was conducted by the senior author. Also to determine the difference between the sides, foot foot "rock paper scissors" and floating toes were observed and examined simultaneously for comparison.

All statistical analyses were performed using the JMP Pro 15 software (SAS Institute, Cary, NC, USA). The relationship between foot "rock paper scissors" and floating toes was analysed using the chi-square test. Results were considered statistically significant if the *P* values were less than 0.05.

Results

A total of 91 baseball players (mean age, 15.1 ± 1.4 years; range, 12–17 years, all men) met the inclusion criteria and were retrospectively evaluated. Sixty players were from high school and 31 players belonged to junior high school. Of the 91 players, 39 players were diagnosed with a disabled throwing shoulder, and 52 players were diagnosed with a disabled throwing elbow. There were no players with duplicate of two or more disorders or players with combined cases of disabled throwing shoulder and elbow. The specific diagnoses of the injuries are presented in Table 1. Among the posterior elbow injury, 3 were diagnosed as posterior elbow impingement and the other was a case of delayed closure of the olecranon epiphyseal plate. Both of the osteochondritis dissecans were cases occurring in the capitellum of the humerus. We diagnosed foot function impairment in 83 players (91%), and floating toes in 74 players (81%). Among the players with

Table 1. Individual player diagnosis. SLAP: superior labrum anterior and posterior.

Disabled throwing shoulder	Internal impingement	31
	Subacromial impingement	5
	SLAP lesion	1
	Thoracic outlet syndrome	2
Disabled throwing elbow	Medial elbow injury	44
	Posterior elbow injury	4
	Osteochondritis dissecans	2
	Ulnar nerve disorders	2

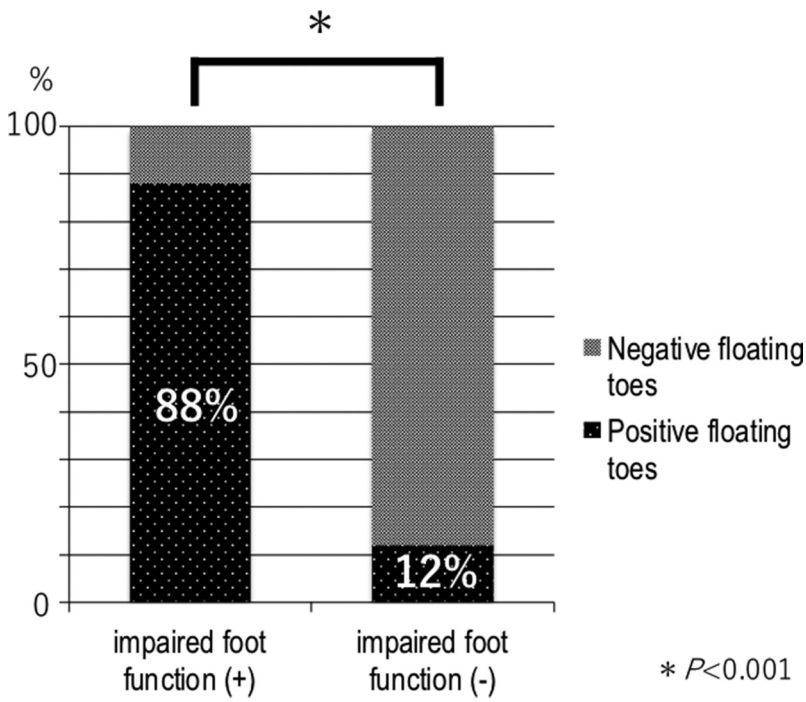

Figure 3. Relationship between floating toes and impaired foot function.

impaired foot function, 73 players (88%) also had floating toes, which the prevalence was significantly greater than those without impaired foot function (12%) ($P < 0.001$; Figure 3). Among the players with floating toes, 73 players (99%) had impaired foot function, which the prevalence was significantly greater than those without floating toes (59%) ($P < 0.001$; Figure 4).

Discussion

Our results revealed that the players with disabled throwing shoulder and elbow had high rates of impaired foot function and floating toes. The human foot has a vital role in supporting body weight, maintaining balance, stabilizing the body, and creating propulsive force to push the body forward (Fukuyama & Maruyama, 2011). The toes play an important role, not only in maintaining the body's balance while standing, and initiating

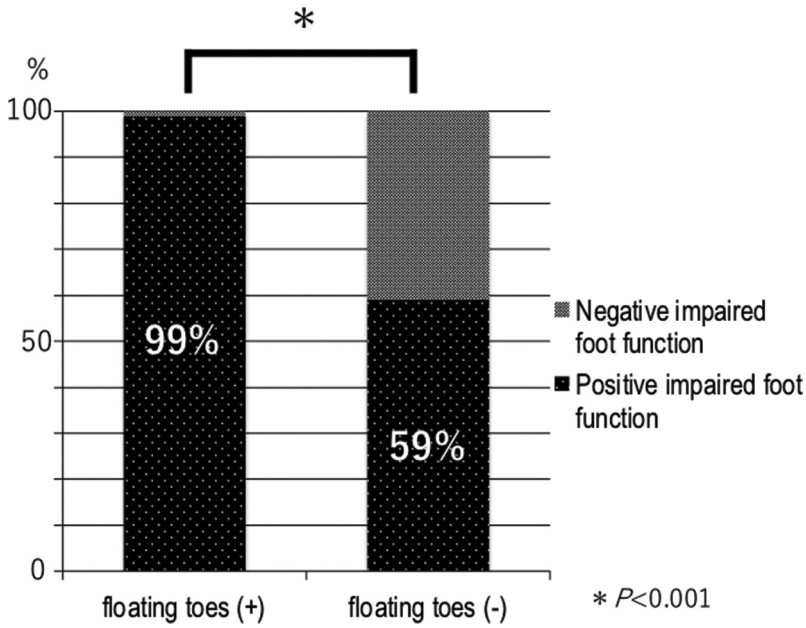

Figure 4. Relationship between impaired foot function and floating toes.

walking movements (Chou et al., 2009), but also through their function as both sensory and effector organs in maintaining posture and ensuring stability during movement (Hasegawa et al., 2006). During throwing motion, the lower extremities play a vital role and proper positioning and strength of both legs are needed to help provide a stable base for the kinetic chain and the remainder of the throwing motion (Chu et al., 2016). The pitchers also need to keep their centre of gravity over their back leg to generate the maximum momentum at the time of initiating forward motion. If the pitcher's centre of gravity is disrupted and the momentum falls forward prematurely, the kinetic chain will be disrupted and a greater force will be required to throw the ball at top velocity (Burkhart, Morgan, Kibler et al., 2003b). From these standpoints, players with impaired foot function may have a dynamic balance deficit of the kinetic chain affecting the execution of the proper pitching motion, eventually resulting in injuries to the shoulder or elbow.

Subjects with floating toes have decreased flexor toe muscle strength compared to those without floating toes and fair worse in foot "rock paper scissors" movements, especially in flexing all the toes to create "rock" (Fukuyama et al., 2009). In cases of subjects with floating toes, toe-grip strength and capability to shift the centre of gravity forward declines due to weakening of the toe flexors, leading to decline in the ability to adjust motion and process the sensory input involved in executing accurate motion (Fukuyama et al., 2009). With regard to sensory input, floating toes are also related to disruption of information input from the floor, due to reduction in the capacity and degeneration of mechanoreceptors (Fukuyama & Maruyama, 2011). Indeed, it is reported that people with floating toes have a tendency towards poorer postural control and dynamic balance (Hasegawa et al., 2013). Hannon et al. (2014) have revealed that baseball players with an ulnar collateral ligament tear had declined

stance and lead leg balance at the time of injury, with significant improvement of these measurements after reconstruction. This improvement in balance may be a result of improvement from the post-operative rehabilitation. Nevertheless, it is an indication that there is a relationship between lower extremity balance and injuries to the shoulder or elbow. Our results and the previously reported data, suggest that floating toes may also have a relationship with dynamic balance deficit, and may relate to disabled throwing shoulder or elbow, similar to the impaired foot function.

The prevalence of floating toes is reported to be around 40% (Saito et al., 2019; Tasaka et al. 2016), and increases in the younger generation (Araki et al., 2015; Harada, 2002). From our results, the prevalence among the players with disabled throwing shoulder or elbow was higher than that in the general population. Physique, physical fitness, balance ability, heel load, shoes, exercise quantity, and usage frequency of toes have been advocated as factors related to floating toes (Harada, 2002; Matsuda et al., 2013; Uchida et al., 2002). In fact, it has been reported that children who played outdoors have significantly lower incidence of floating toes compared to those who play indoors (Onuki et al., 2005). There are other reports regarding footwear as an environmental and lifestyle factor for the cause of floating toes (Yahagi et al., 2004). Wearing loose shoes or slippers tends to make toes spread during walking to pursue stability, and cause them to stretch to prevent the shoes or slippers from coming off (Yahagi et al., 2004). These compensatory movements lead to heel load or displacement of the centre of gravity to a posterior position, which is one of the primary reasons for floating toes (Harada, 2002; Uchida et al., 2002). Matsuda et al. (2013) reported that posterior foot pressure ratio was significantly larger in children with over two floating toes than in children without floating toes or with one floating toe. Thus, children with floating toes may have a posteriorly shifted centre of gravity and decreased stability in standing posture, resulting in abnormal dynamic balance and a negative effect on postural maintenance. Further studies may be required to determine the relationship between posterior weight shift during standing posture, motor development, and environmental/lifestyle factors, which may lead to floating toes.

Our study has several limitations in this study. First, there were no study controls. As we conducted the study retrospectively on the players visiting the outpatient clinic, evaluating players without having a disabled throwing shoulder or elbow was impossible. Evaluating non-injured controls will be needed to assess if impaired foot function and floating toes relate to the injury. Second, the results of the floating toes did not consider the dominant hand side. As baseball is a sport conducted unilaterally, the existence of the differences in the degree of impairment of foot function and floating toes between the sides may have affected the injury. However, there were no subjects who had different results between the two sides, and as a result, we did not define the results by side. Third, we did not analyse floating toes separetely. It has been reported that the float frequently involves the 4th and 5th toes (Uchida et al., 2002). Analysing the floating toes specifically may be needed in future studies. Fourth, we were not able to determine if the impaired foot function and floating toes were the cause of the injury. Further studies are required determine whether improving the foot function and floating toes will have a positive effect on the occurrence of disabled throwing shoulders and elbows. Fifth, footwear was not taken into account, although it could be one of the

causes of floating toes (Yahagi et al., 2004). As baseball cleats are not uniformly defined and can be switched frequently during the season, we assumed that analysing by baseball cleats was impractical. However, taking footwear into consideration in the future studies maybe needed.

Conclusions

Baseball players with disabled throwing shoulders and elbows showed high rates of impaired foot function and floating toes. This foot pathology may have a relationship with the occurrence of disabled throwing shoulder and/or elbow.

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